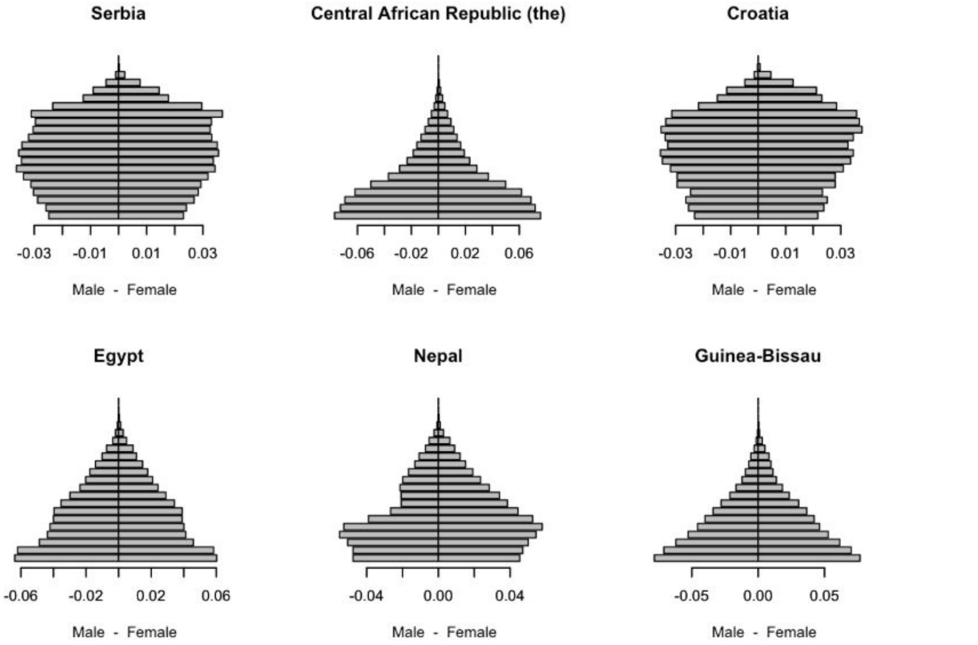
Assignment 1

Density Estimation and Clustering

Sofia Nightingale, Ferran Estrella, Alessandro Valls

Part 1

Population pyramids dataset



Part 1: Kernel Density Estimation

We want the best bandwidth: $\boldsymbol{h_{LCV}}$

$$h_{LCV} = \arg\max_{n} \sum_{i=1}^{n} \log \hat{f}_{h,(-i)}(x_i)$$

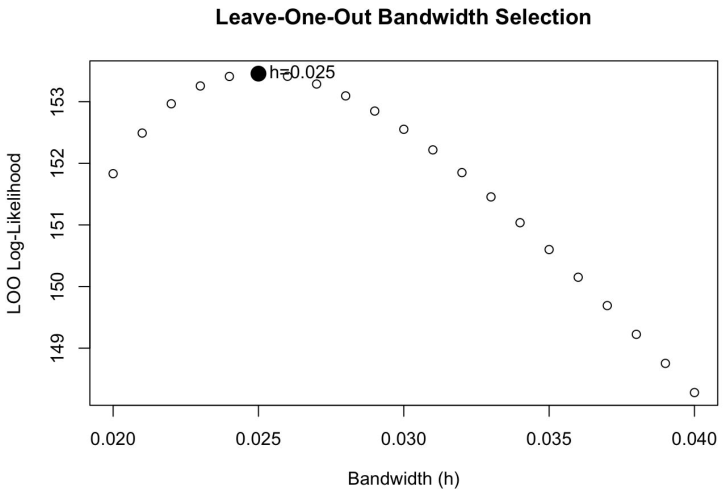
$$\hat{f}_{h,(-i)}(x_i) = \frac{n}{n-1}(\hat{f}_h(x_i) - \frac{K(0)}{nh})$$

$$\Downarrow$$

```
kx <- density(x, bw=h, n=2048)
kx_f <- approxfun(x=kx$x, y=kx$y, method='linear', rule=2)
fESR <- (n/(n-1))*(kx_f(x)-(dnorm(0)/(n*h)))
```

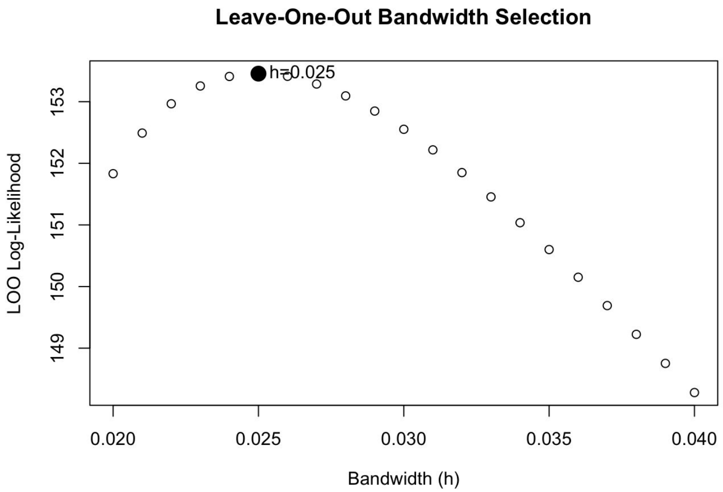
Part 1: Kernel Density Estimation

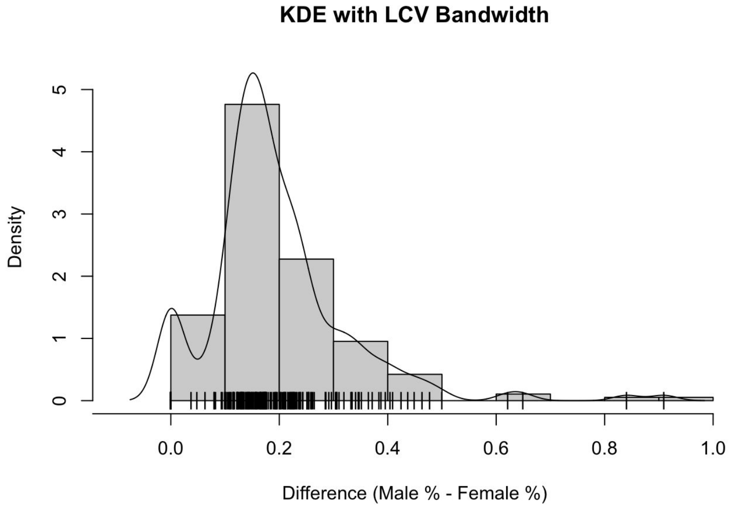
$h_{LCV} = 0.025$



Part 1: Kernel Density Estimation

$h_{LCV} = 0.025$





Part 1: Clustering

Hierarchical VS K-medoids

Part 1: Clustering

Best number of clusters: k = 3

Hierarchical

Average Silhouette Criterion

K-medoids

Calinski and Harabasz index

Part 1: Clustering

Best number of clusters: k = 3

Hierarchical

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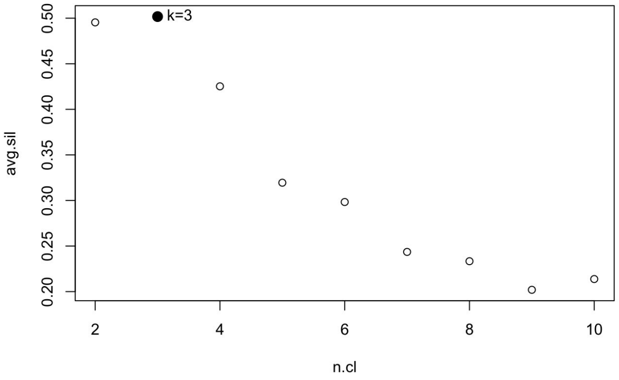
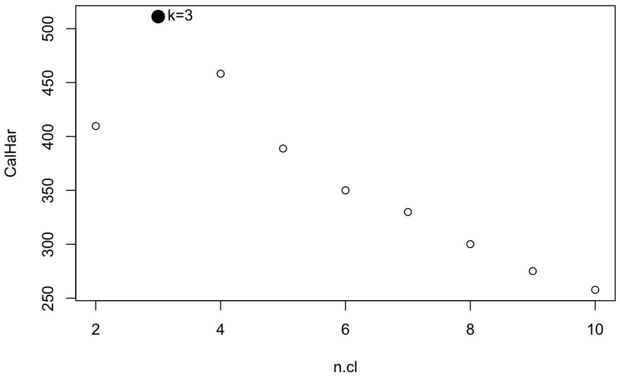


K-medoids

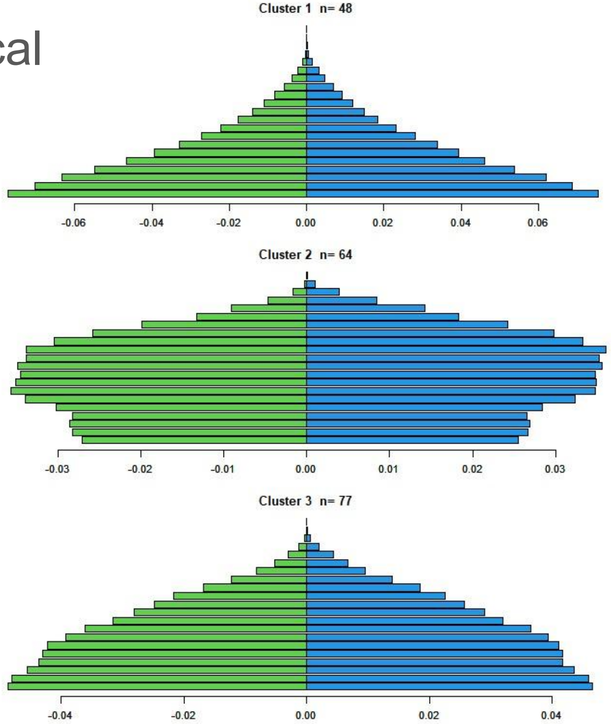
Calinski and Harabasz index



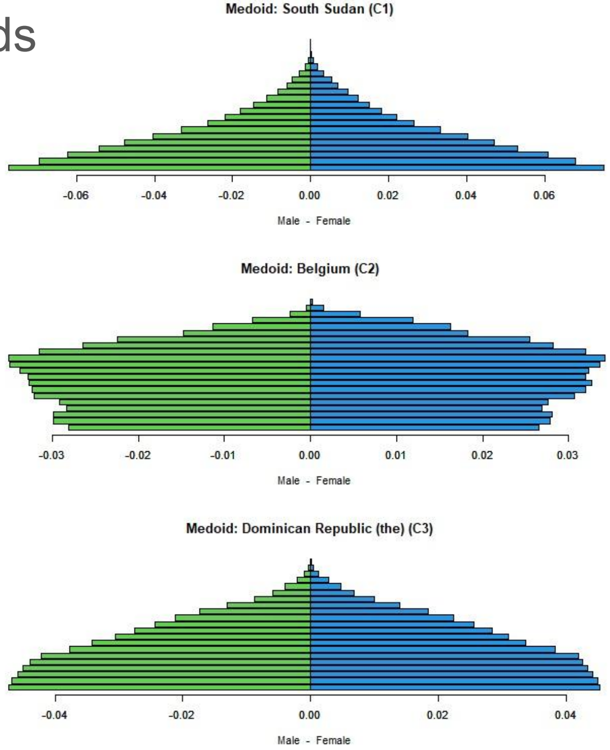
Part 1: Clustering

Average population pyramids

Hierarchical



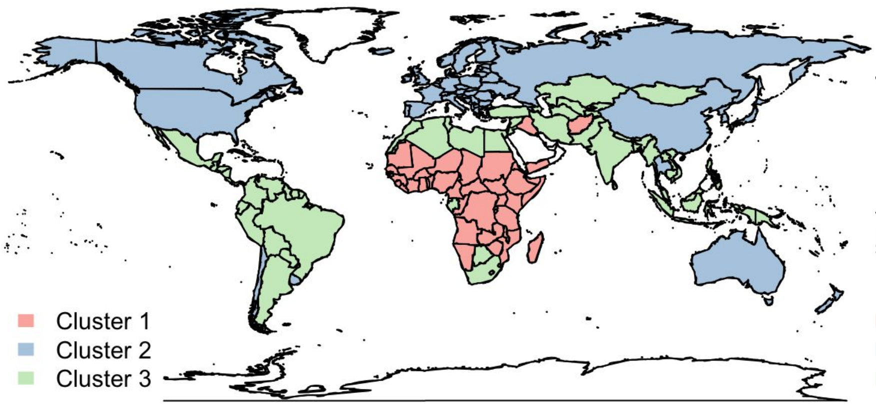
K-Medoids



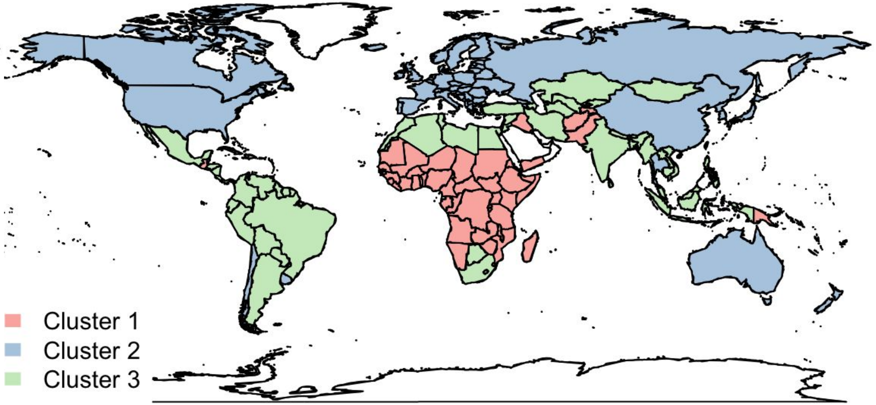
Part 1: Clustering

World maps

Hierarchical

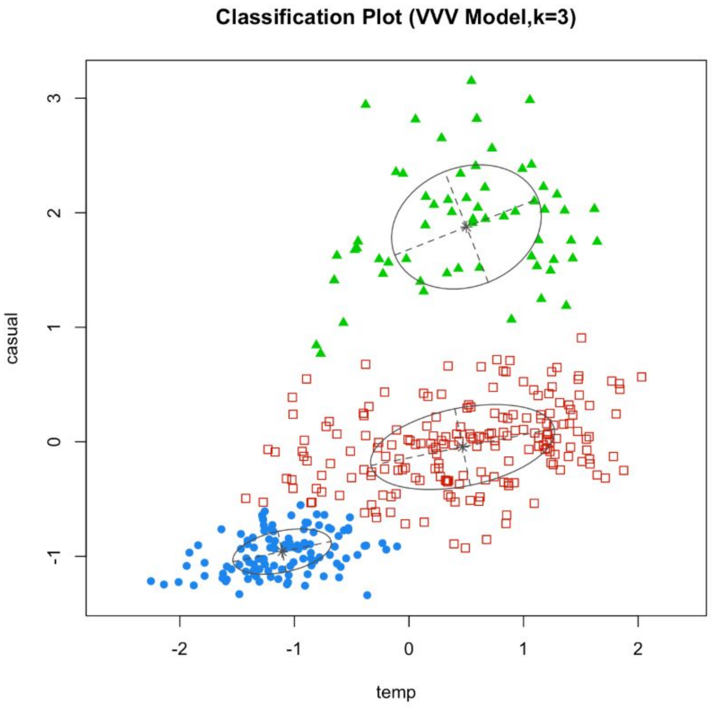


K-medoids



Part 2

Bike-sharing dataset

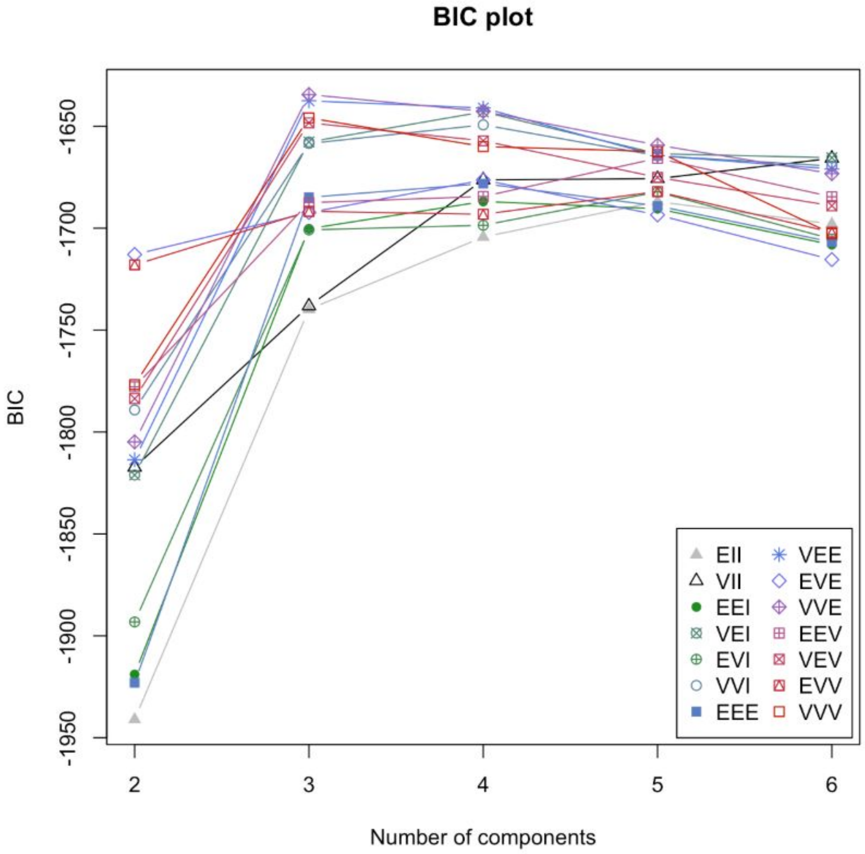


Part 2: Gaussian Mixture Model (GMM)

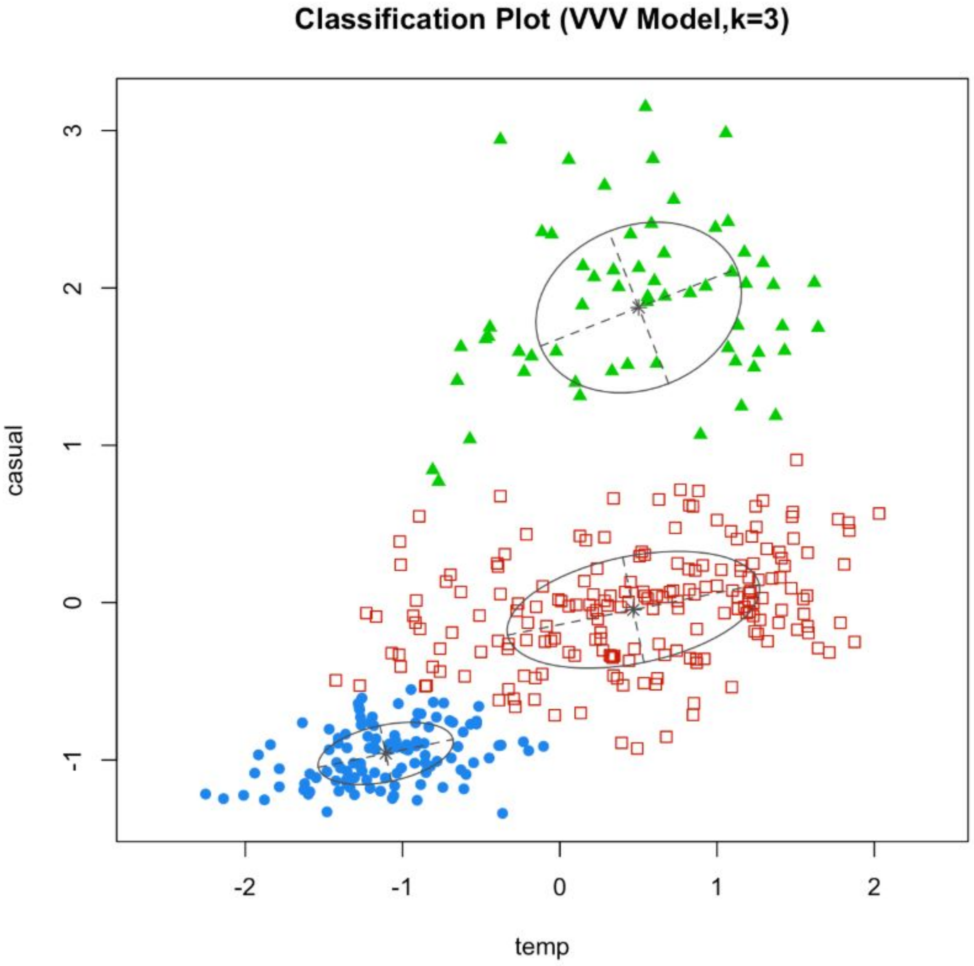
- 2D data: (temp, casual)
- GMM as clustering method
- VVV modelling
- Software: Mclust
- Number of clusters maximizing

$$\mathrm{BIC} = 2\log L_{\max} - p\log n$$

is **k=3**



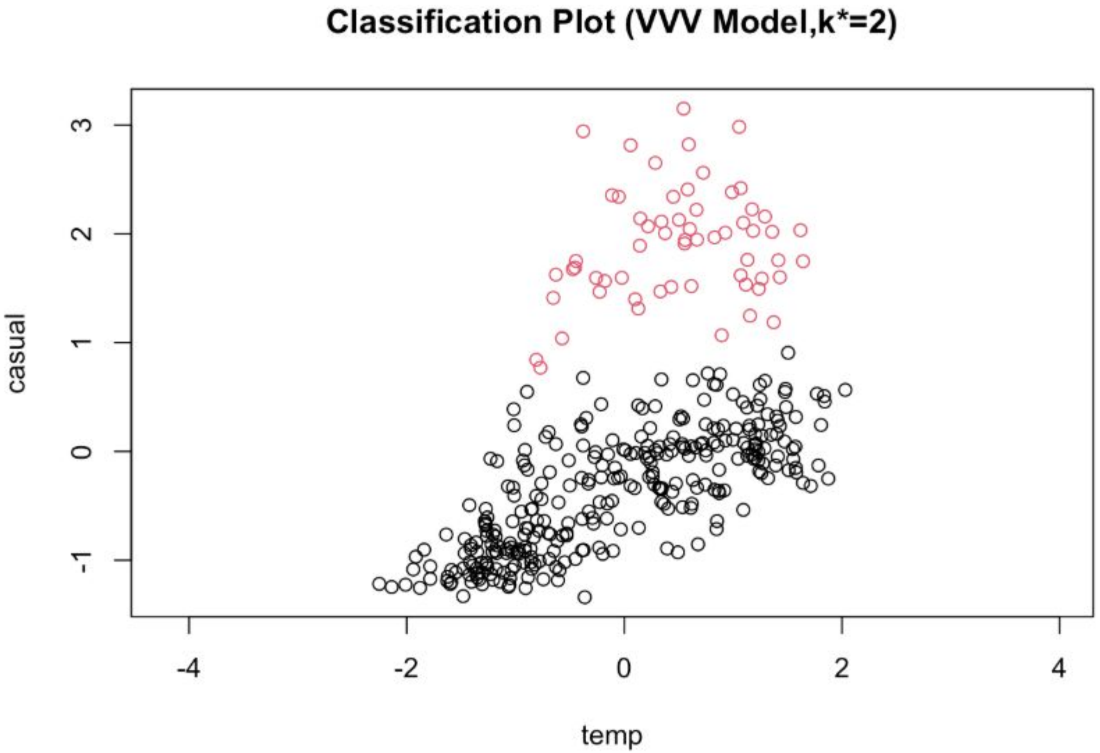
Part 2: GMM plot



Part 2: Cluster Merging

- Software: mergenormals (method="bhat")

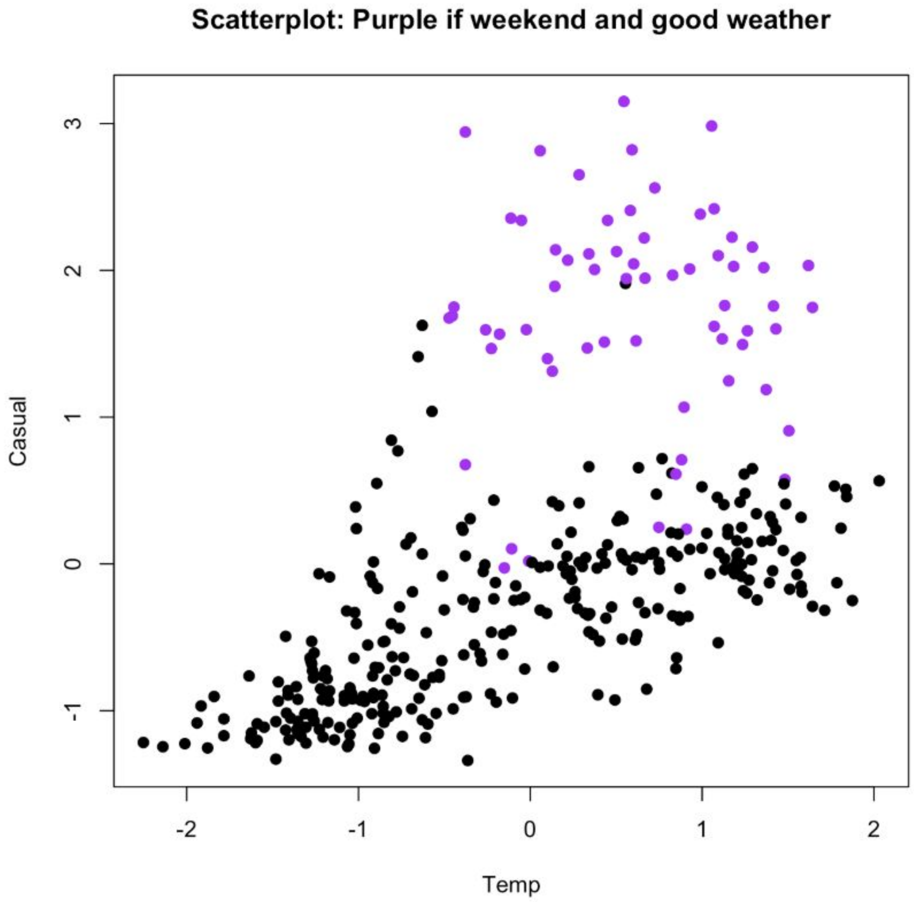
- Merging of the two more dense clusters



Part 2: Interpretation

- Less dense cluster: Weekend/holiday with good weather

- Unusually high casual activity, but less probable (few days of the year)



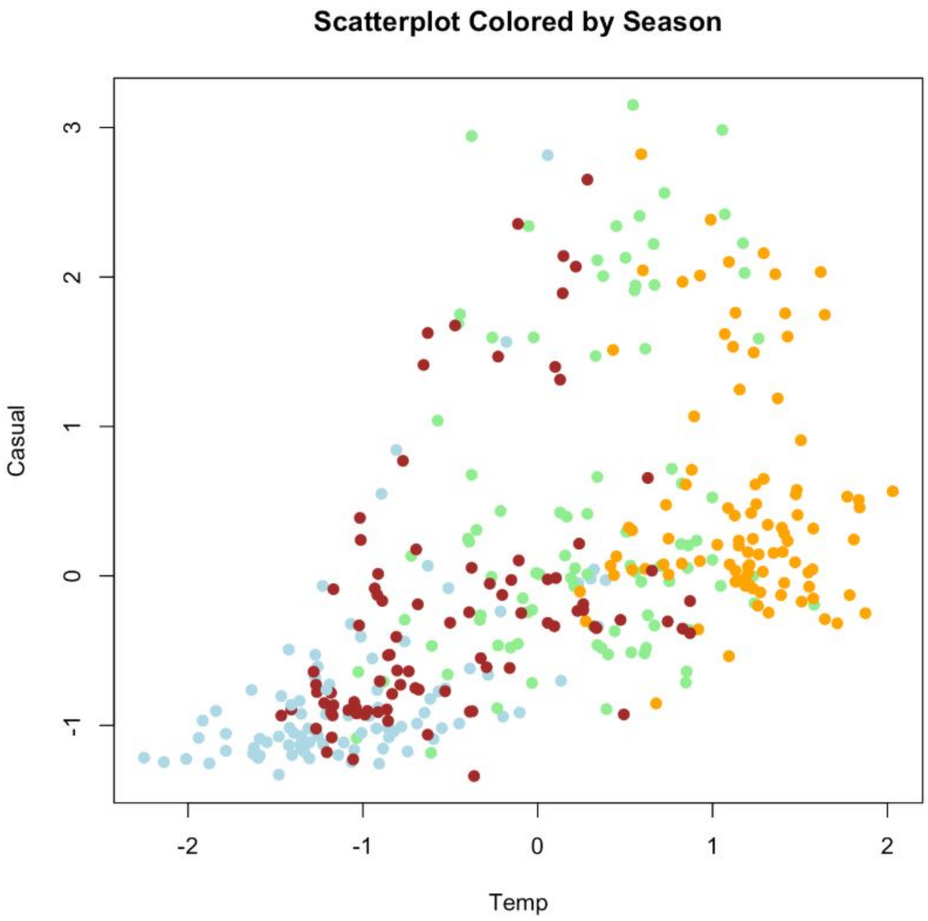
Part 2: Interpretation

- **Dense cluster, low temperatures:**
  Cold days (winter, fall, …)

  Low casual activity

- **Dense cluster, high temperatures:**
  Hot days (summer, spring, …)

  Low casual activity, but higher



Assignment 2

PCA and MDS

Sofia Nightingale, Ferran Estrella, Alessandro Valls

Part 1

Principal Component Analysis (PCA)

Part 1: PCA

- Data matrix X: Population proportions for age ranges and gender, for **n** countries (population pyramids)
- PCs:

$$\mathbf{Y} = (\mathbf{y_1}, \ldots, \mathbf{y_p}) = (\mathbf{X} - \mathbf{1}_n \mathbf{m}')V$$

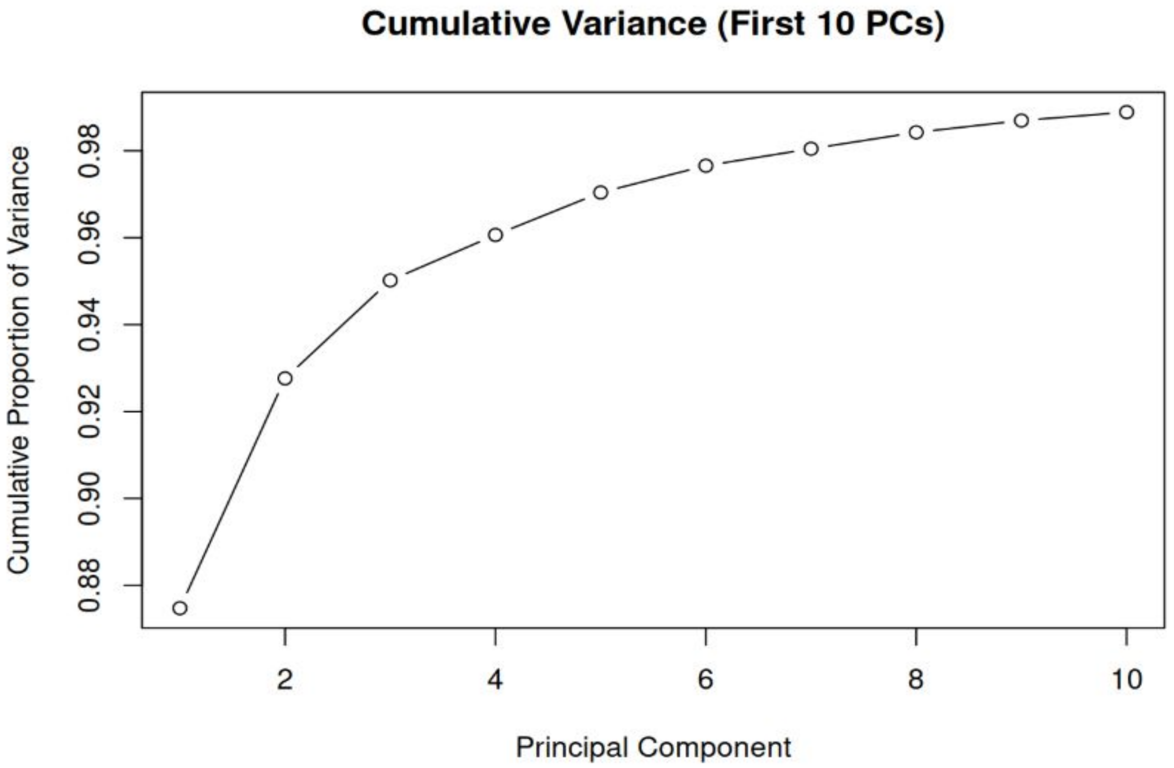
- V evecs of evals λ_k of variance matrix:

$$\mathbf{S} = \frac{1}{n-1}(\mathbf{X} - \mathbf{1}_n \mathbf{m}')'(\mathbf{X} - \mathbf{1}_n \mathbf{m}') \qquad \mathrm{var}(\mathbf{y_k}) = \lambda_k$$

- Software: princomp

Part 1: PCA

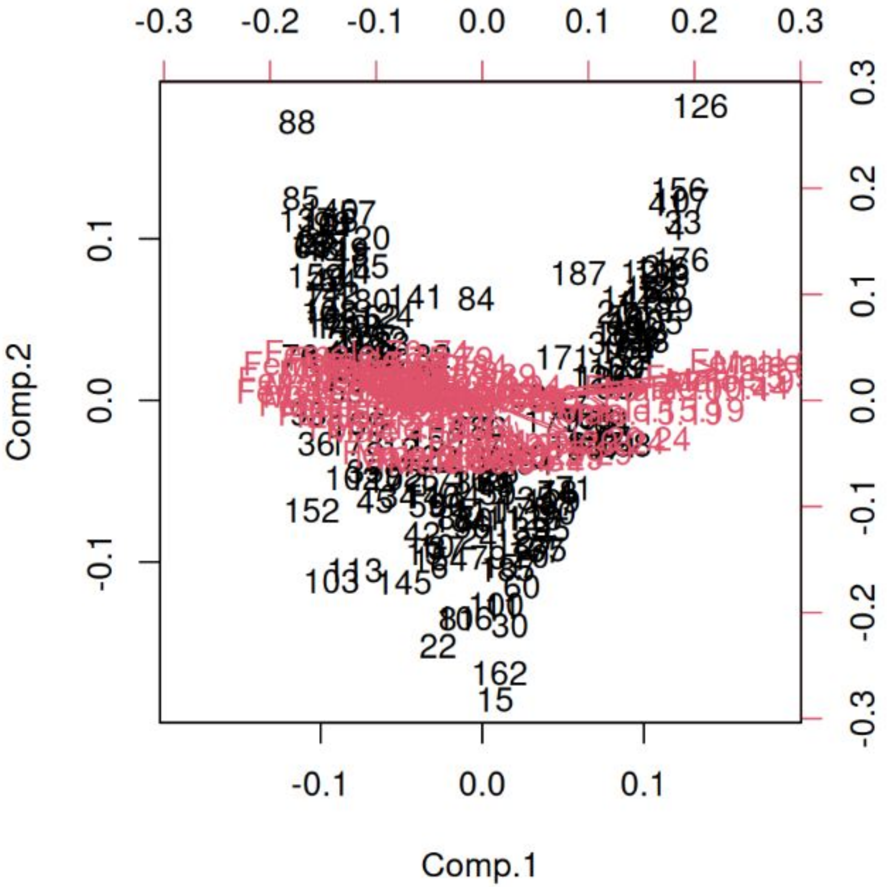
Variance proportion:
var(y)/Tr(S)

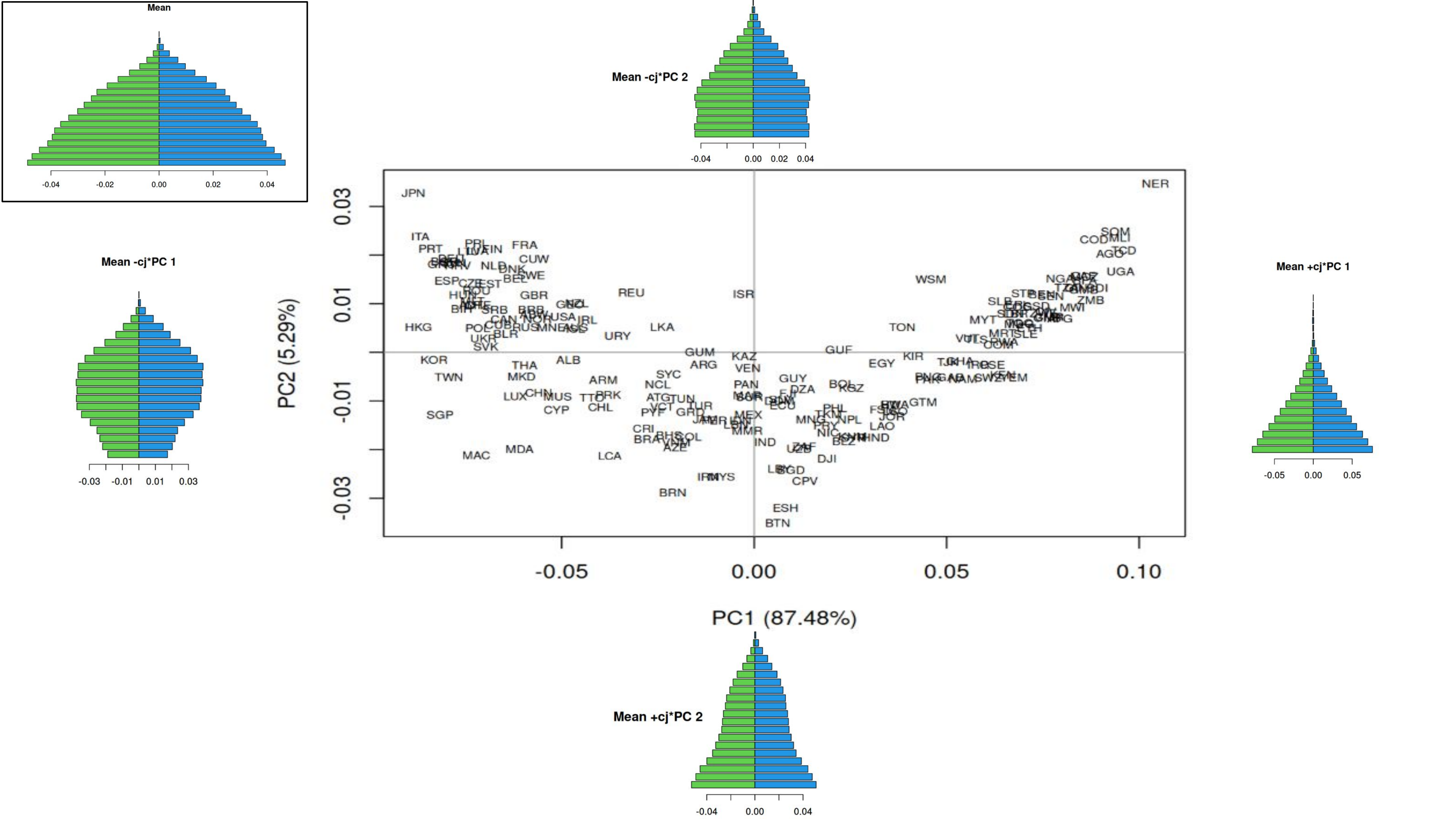


Part 1: Biplot

-Black: First and second PC scores→ V shape→Possible unidimensional non-linear dimensionality reduction

-Pink: First and second PC loadings of the variables → Aligned with PC1

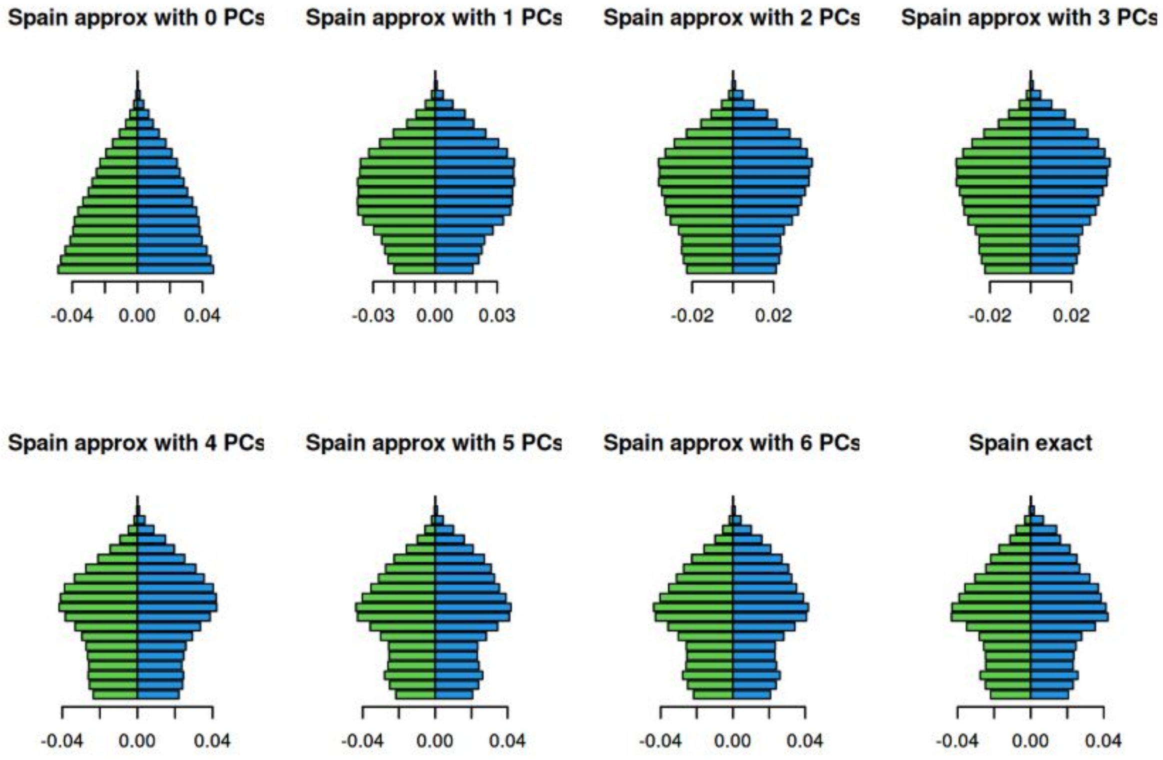




Mean
-0.04
-0.02
0.00
0.02
0.04
Mean -cj*PC 2
Mean -cj*PC 1
Mean +cj*PC 1
Mean +cj*PC 2
-0.03
-0.01
0.01
0.03
-0.05
0.05
0.10
PC1 (87.48%)
PC2 (5.29%)
JPN
NER
ITA
PRT
FRA
CUW
SOM
TCD
AGO
UGA
NLD
ESP
WSM
HUN
GBR
REU
ISR
ZMB
MWI
HKG
LKA
TON
URY
POL
UKR
SVK
GUM
GUF
KOR
THA
ALB
KAZ
ARG
EGY
KIR
TWN
MKD
ARM
SYC
VEN
GUY
NCL
PAN
BOL
LUX
MUS
CYP
CHL
GTM
SGP
PHL
NPL
LAO
MEX
CRI
MMR
IND
HND
MAC
MDA
LCA
DJI
CPV
BRN
ESH
BTN

Part 1: Data Dimensionality Reduction

$$\mathbf{x_i^{(k)}} = \mathbf{m} + \sum_{j=1}^{k} y_{ij}\mathbf{v_j}$$



Part 2

Multidimensional scaling (MDS)

Part 2: MDS

Dimensionality reduction based on inter-individual distances

Classical

- preserves the exact absolute demographic distances

Non-metric

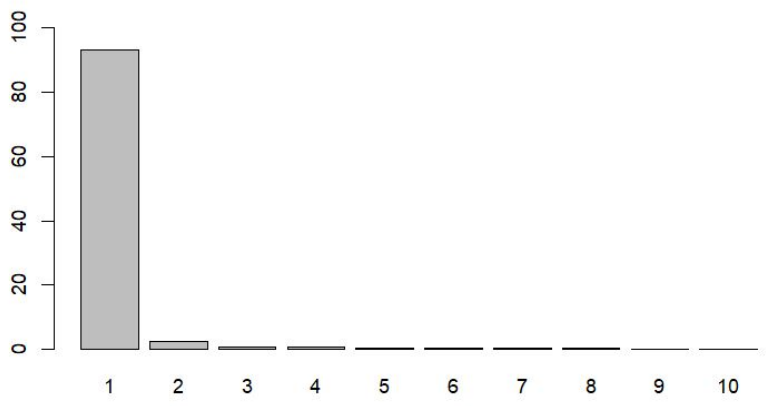
- uses an iterative algorithm (taking Classical as the first guess)
- preserves rank order of distances

Part 2: MDS

Methods

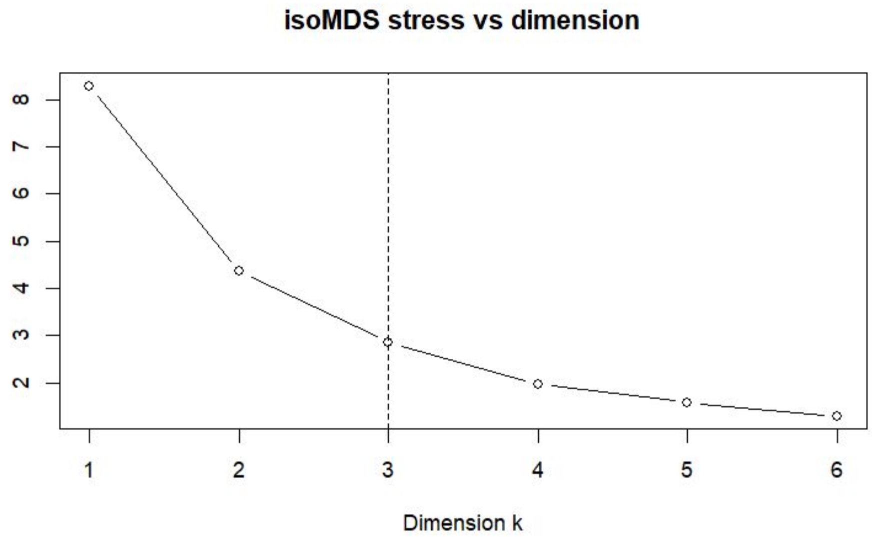
Classical

cmdscale



Non-metric

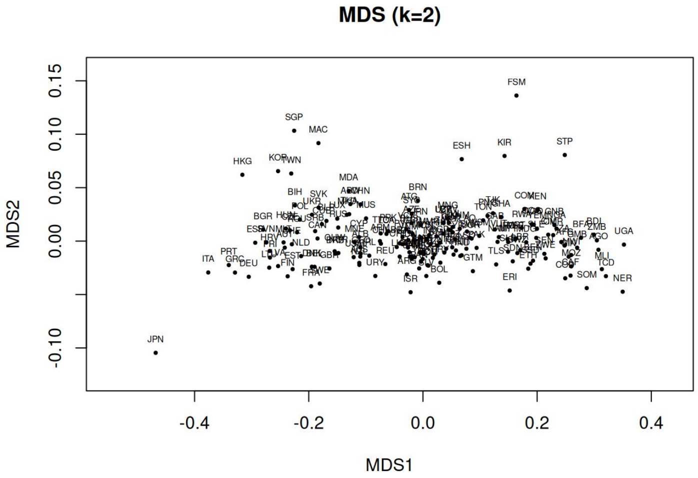
isoMDS



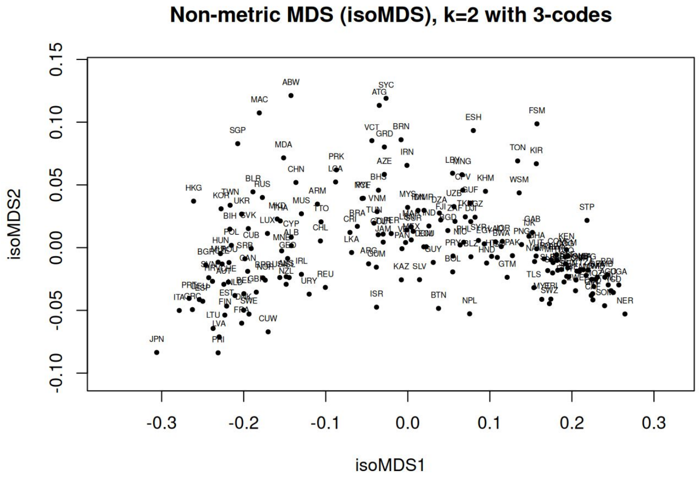
Part 2: MDS

2D data representation

Classical



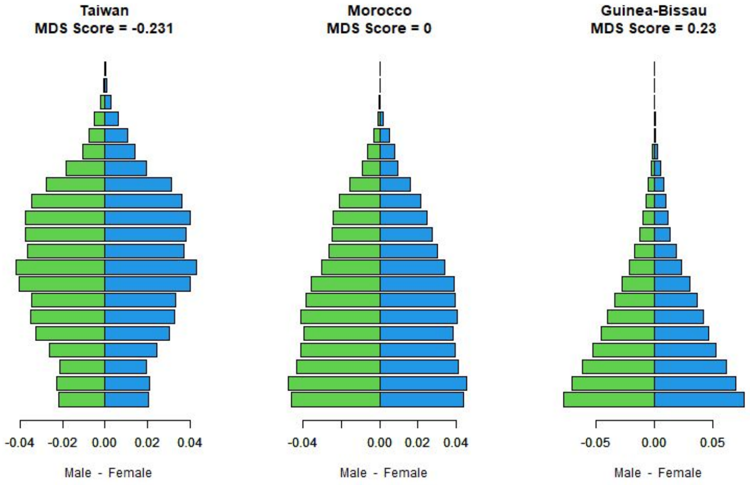
Non-metric



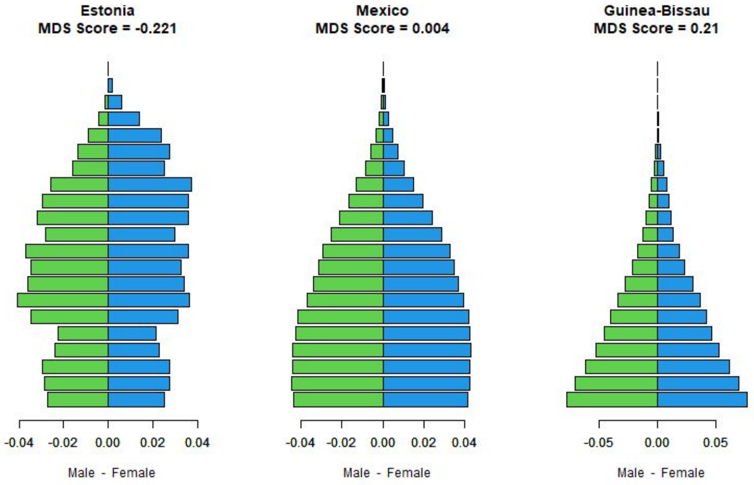
Part 2: MDS

Interpretation: Dimension 1

Classical



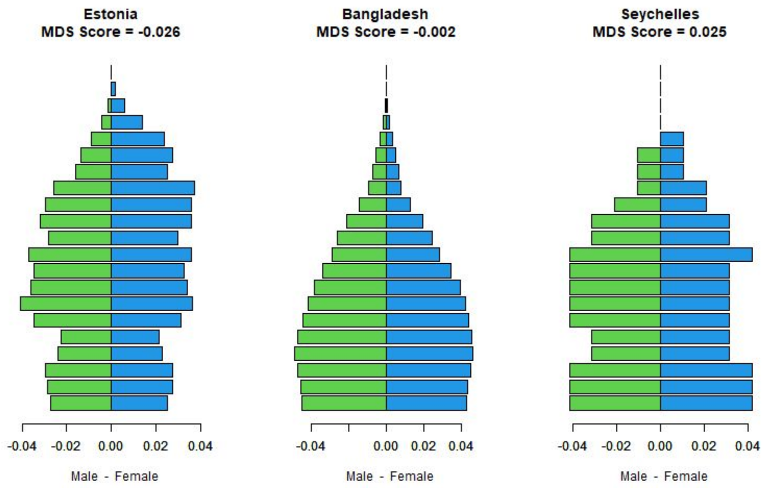
Non-metric



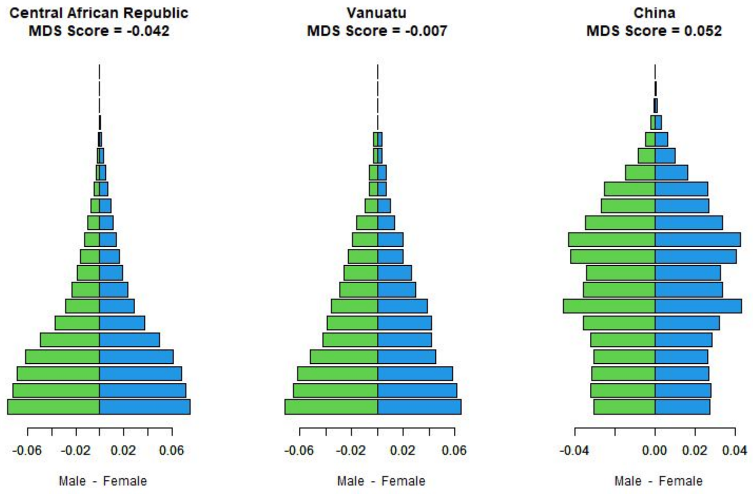
Part 2: MDS

Interpretation: Dimension 2

Classical



Non-metric

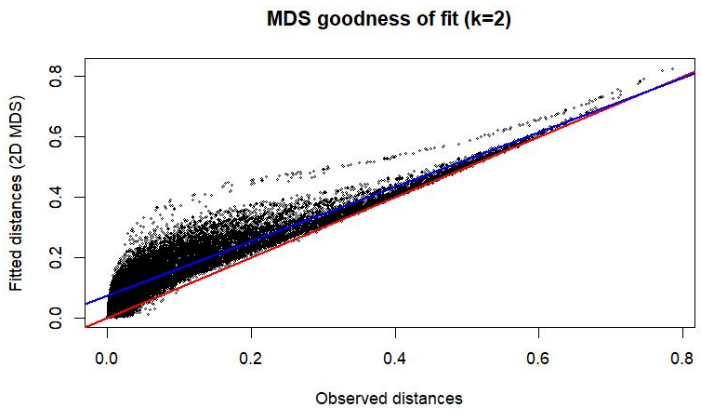


Part 2: MDS

Quality of fit

Classical

Goodness of fit plot



Non-metric

Stress

```
STRESS_NM(D,X,f) = 4.35
```

Part 3

PCA & MDS world maps

Part 3: PCA vs MDS

World map plot

PCA

MDS